

REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie



REPUBLIC OF CAMEROUN
Peace-Work-Fatherland



Towards a reference framework for agroforestry cocoa in Cameroon

Development process and definition criteria



Mise en œuvre par



En coopération avec



1. Context and justification

Cameroon mainly produces its cocoa using complex agroforestry systems (AFS) that provides numerous environmental benefits and contributes to the socio-economic resilience of rural households.

Faced with new requirements of international markets, notably the European Union Regulation on deforestation and forest degradation (EUDR), which considers cocoa farming in agroforestry to be an agricultural practice, promoting the specificities of cocoa farming in Central Africa become strategic. Cameroon, with its strong heritage of complex agroforestry systems, has therefore opted to establish a national reference framework for its agroforestry cocoa, permitting:

- To objectively qualify cocoa-based agroforestry systems,
- To support a regulatory and as well a commercial recognition of cocoa produced under broad shade,
- To maintain biodiversity, forest cover and ecosystem services in these complex agroforestry systems
- To contribute to the resilience of these production systems and rural households.

2. Objectives and approach to co-constructing the reference framework

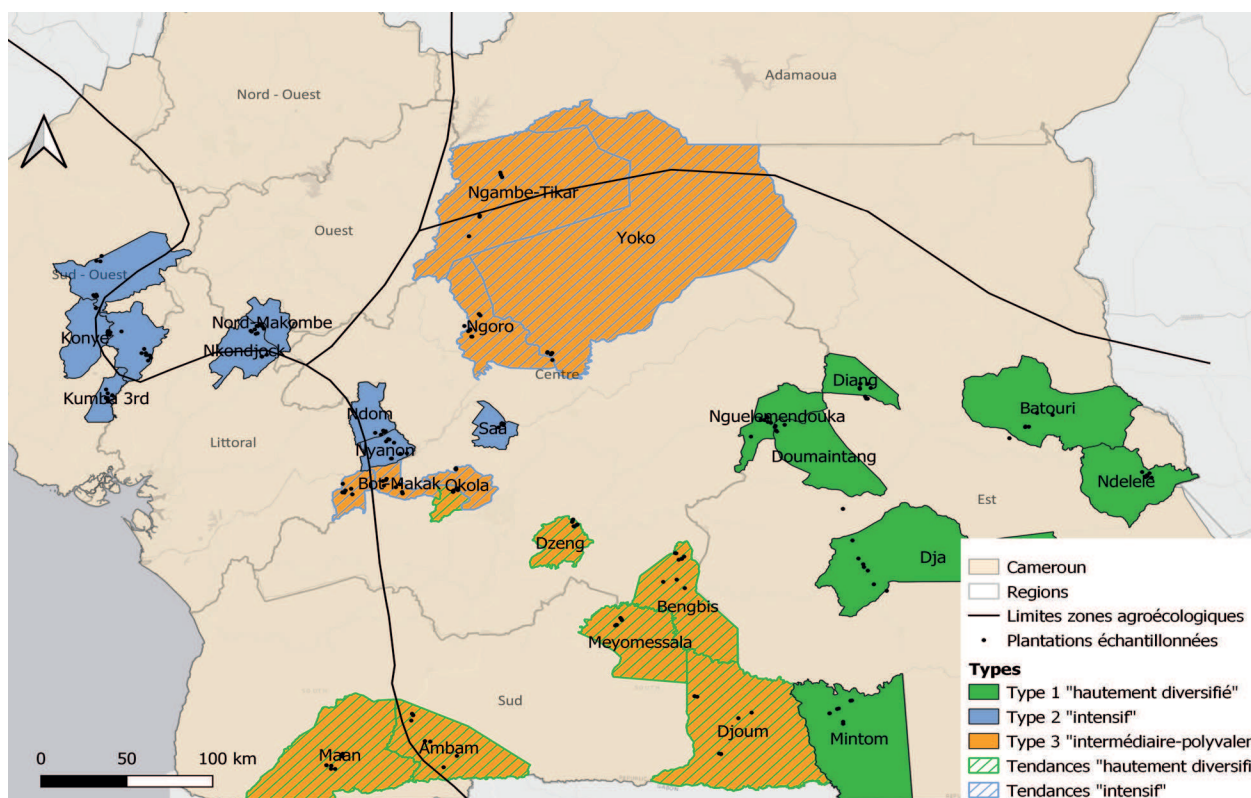
Develop a consensual technical framework for characterizing and promoting cocoa-based agroforestry systems in Cameroon, tapping from:

- a solid scientific basis ;
- a multi-stakeholder participatory approach over a period of two years;
- a political will for positive differentiation.

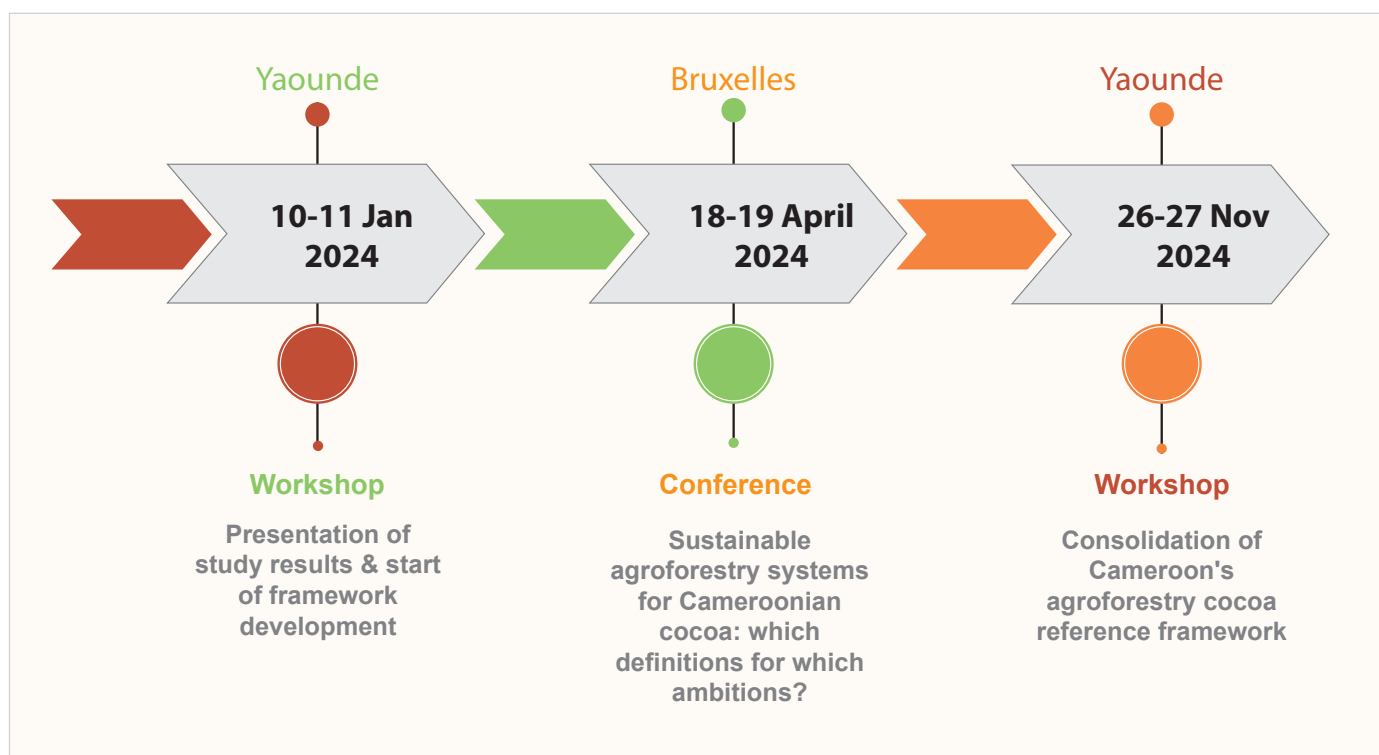
2.1 Scientific reference study

From April to December 2023, a study on the characterization of cocoa-based AFS was jointly conducted by the Centre for International Cooperation in Agricultural Research for Development (CIRAD) and the Interprofessional Cocoa and Coffee Council (CICC), with the support of the European Union's Sustainable Cocoa Programme, implemented by GIZ. The characterization was carried out by measuring 17 structural, ecological and economic criteria in 223 cocoa plantations in five regions and three agroecological zones.

The results revealed three types of AFS, namely: (i) the «**highly diversified**» system consisting of cocoa plantations having the highest densities and diversity of associated forest trees; (ii)



Criteria		«Diversified» (25%)			«Intensive» (35%)			«Multipurpose» (40%)		
		Minimum	Median	Average	Minimum	Median	Average	Minimum	Median	Average
1	Age of cocoa farm (years)	5	12	15	13.5	38	40	6	13	19
2	Yield/year (kg)	123	180	234	571	774	700	226	434	443
3	Phytosanitary cost (CFA francs/ha/yr)	5 000	7 917	6 242	40 000	60 000	78 575	8 500	31 667	34 996
4	Number of interventions	2	5	5	7	8	10	5	9	9
5	Number of uses of forest species	6	8	9	7	9	9	8	15	14
6	Number of cocoa trees/ha	588	750	759	675	913	944	675	900	963
7	Number of fruit trees/ha	2	23	19	2	13	16	6	21	23
8	Number of forest trees/ha	79	117	116	33	38	41	29	71	66
9	Number of associated trees/ha	127	138	149	50	54	59	66	87	89
10	Number of associated species per 1/4 ha	30	33	36	12	13	14	16	21	21
11	Basal cover of cocoa trees (m ² /ha)	1.44	5.12	5.11	8.86	16.76	15.09	2.79	6.83	8.76
12	Basal cover of fruit trees (m ² /ha)	0.01	0.63	0.77	0.2	0.38	0.64	0.09	0.93	1.26
13	Basal cover of cocoa trees (m ² /ha)	9.23	22.45	20.03	3.39	9.7	11.53	5.89	11.36	13.59
14	Basal cover of associated trees (m ² /ha)	20.51	24.36	24.47	3.76	10.42	11.53	6.84	12.25	13.92
15	Total basal cover (m ² /ha)	27.4	29.57	30.41	16.56	35.24	29.38	10.67	20.77	22.17
16	Shannon index	2.45	2.59	2.59	1.87	1.92	1.99	1.95	2.29	2.24



the ‘**intensive**’ system, consisting of cocoa plantations having the highest values in terms of age, yield and basal area of cocoa trees as well as the highest levels of intensification in terms of labour and inputs; and finally (iii) the «**multipurpose**» system consisting of cocoa plantations with a large number of fruit trees per hectare and uses of associated forest species.

2.2 Multi-stakeholder co-construction process

Based on the results of this study, a multi-stakeholder co-construction process was carried out during 2024 with the aim of defining a reference framework of the complex cocoa-based agroforestry system that will be operational and easy to promote in Cameroon and on international markets.

In addition to consultations, interviews and discussions with key stakeholders, the process was significantly marked by the holding of three workshops in which different categories of stakeholders took part: public administrations, the private sector, researchers, NGOs, producer organisations, technical and financial partners, and foreign delegations (from other cocoa-producing countries).

A comparison of the study’s results with other sustainable cocoa standards, particularly those in West Africa, highlighted the need to define a complex agroforestry cocoa standard, at least for Cameroon. Indeed, while in West Africa the objective is to reintroduce (simple) agroforestry into quasi-monocultural production systems that have been built at the expense of forests, in Cameroon, and in Central Africa more generally, the stake is to maintain – or even expand – the traditional complex agroforestry systems that have developed while preserving most of the ecological attributes of a degraded forest and ensuring ecological and socio-economic resilience.

There is therefore an interest in promoting existing complex agroforestry systems in Cameroon, rather than being tempted by the trajectories observed in West Africa.

3. Criteria and thresholds selected for the Cameroon agroforestry cocoa reference framework

The January 2024 workshop permitted to identify four variables that could be used to classify Cameroonian agroforestry cocoa, namely: (i) the maximum number of cocoa trees per hectare; (ii) the minimum number of associated trees per hectare; (iii) the minimum basal area of trees associated with cocoa trees; (iv) three plant strata. Numerous arguments were presented and discussed during this workshop, but no compromise was reached by the stakeholders, partly due to the technical nature of the debate, but mainly because of its political, economic and social implications.

It was during the workshop to consolidate the reference framework in November 2024 that a consensus was reached on the variables and thresholds to be used to define the reference framework. The debate focused not only on figures, but also on the philosophy of the agroforestry model to be preserved. Four main principles guided the final choices:

Avoid minimising agroforestry systems: researchers insisted that thresholds should not become simplification targets. Setting a minimum value that was too low risked transforming a complex agroforestry system into an impoverished system that would nevertheless be “compliant” on paper. Hence the preference for intervals (e.g. cocoa tree density) and the introduction of structural criteria (strata, basal area, canopy cover) rather than just tree counts.

Maintaining the ecological identity of Cameroonian cocoa AFS: the aim was to preserve the ecological functions of AFS, namely: biodiversity (number of associated and native species), forest structure (number of strata), biomass and carbon storage (basal area), microclimate and water regulation (forest cover) and the maintaining of a forest character (number of forest trees).

Reconciling ecology and profitability: A density of 850–1,100 cocoa trees per hectare is recognised as optimal for yield, but this has been classified as a non-binding recommendation. This choice recognises that cocoa is an economic activity, but some high-performing agroforestry systems have lower densities. This was in order not to exclude traditional systems that are ecologically rich but less intensive. All this is to ensure that the reference framework is not seen as a model for intensification, but as a framework for structural sustainability.

Designing a simple reference framework, both in terms of the number of criteria and how they are monitored in the field

This reasoning led to a consensus on seven binding criteria for this reference framework.

Criterion	Minimum threshold
Number of associated trees/ha	≥ 70
Number of forest trees (≥5 cm DBH)/ha	≥ 45
Number of associated tree species/ha	≥ 12
Number of native tree species/ha	≥ 10
Basal area of associated trees (m ² /ha)	≥ 11
Forest cover	≥ 40%
Number of plant strata	3

An additional but non-binding indicator has also been recommended, namely to project for a number between 850 and 1,100 cocoa trees per hectare.

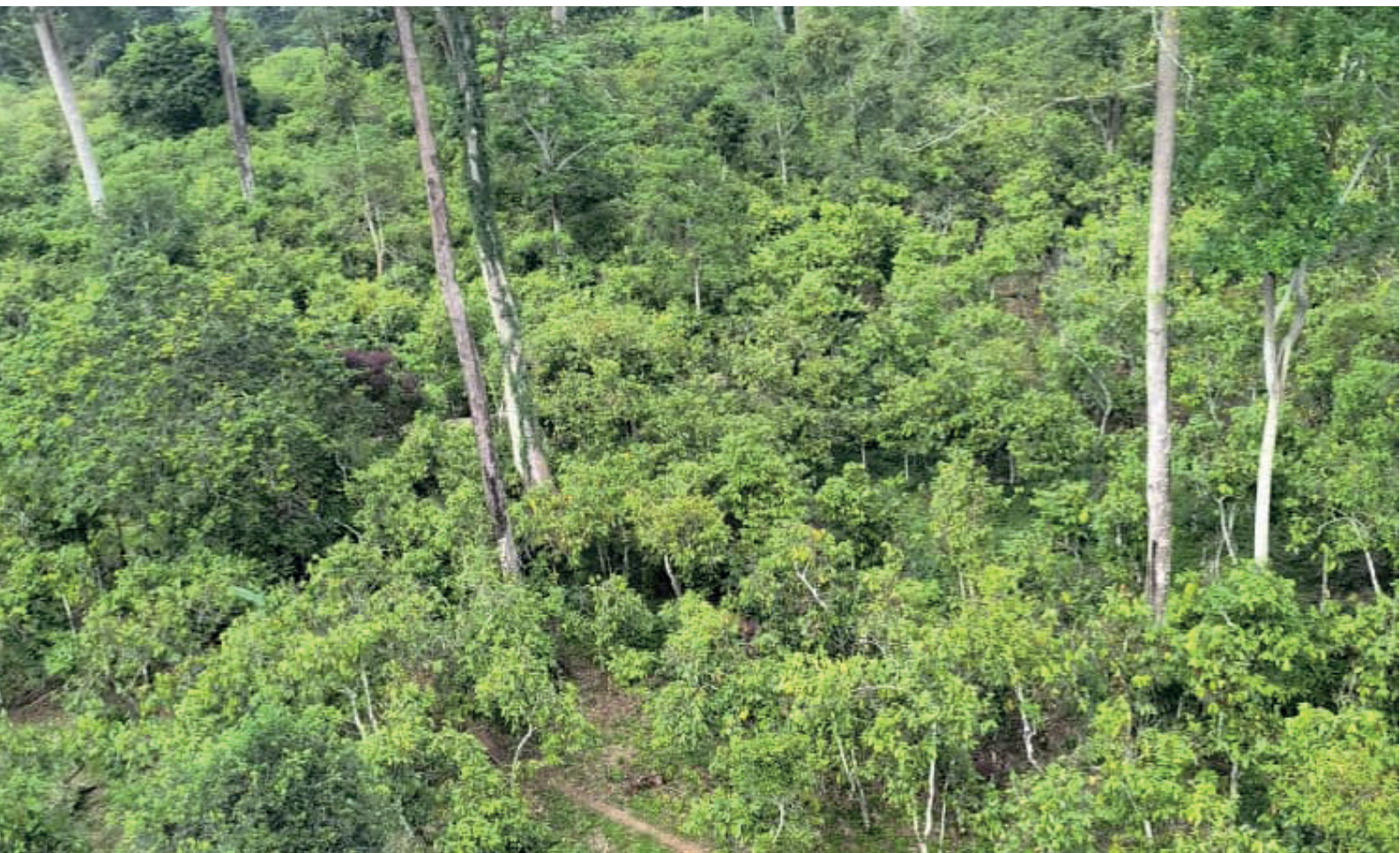
4. Scope of the reference framework

The joint development of this framework can have several scopes, which can also be reconciled:

- Incorporate the agroforestry cocoa reference framework into the reference manual for the production and marketing of sustainable cocoa in Cameroon, or at the regional level;
- Be recognised by an international sustainability standard (Rainforest Alliance, Fairtrade, etc.);
- Contribute to the revision of the EUDR, particularly with regard to the definitions of agroforestry and forest degradation;
- Convince the private sector to promote this agroforestry cocoa, for example by voluntarily adopting this reference framework

5. Outlook

Action	Responsible parties	Target deadline
Inclusion in Cameroon's sustainable cocoa reference manual	EFI, ONCC	2026-2027
Dissemination (manual, training)	GIZ, CICC, researchers	2026-2027
Conducting two complementary studies: (i) Comparative study of the ecosystem performance of AFS and adjacent degraded forests (ii) Presentation and discussion of the study results	CICC, CIRAD, GIZ-SCP	2025-26
Followed by advocacy for the recognition of AFS by the EU		2026-2028
Sub-regional workshop for an agroforestry cocoa reference framework for Central Africa	CIRAD, CICC, GIZ-SCP	March 2026
Advocacy for sub-regional recognition of the reference framework	COMIFAC, CAISTAB, INERA, ECCAS,	June 2026
Scientific publication	CIRAD, CICC	2026



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